MedTrack_DV

Hospital Operations & Patient Analytics Dashboard

Dashboard Storyboard and Interaction Plan

Technology: Python, Pandas, Plotly, Streamlit and Excel

Project Objective

Transform hospital patient and resource datasets into interactive visual insights that help users explore patient outcomes, readmissions, length of stay, costs, bed occupancy and staffing.

Implementation Approach

The project uses Streamlit instead of the Tableau workbook originally specified. The existing application has four pages: Overview, Patient Analytics, Resource Utilization and Data Explorer. This storyboard documents the intended use of those pages and identifies areas where the available data limits the analysis.

Dataset	Records	Purpose
Patient analysis dataset	984	Patient outcomes, readmissions, stay and cost
Resource allocation dataset	200	Bed allocation, occupancy and staffing
Healthcare dataset	500	Additional clinical information collected during preparation

1. Dashboard Storyline

The user begins with an overall snapshot, investigates patient metrics, examines available resources, and finally inspects the underlying data.

Order	Dashboard Page	Question Answered	Main Content
1	Overview	What does the overall patient and resource picture look like?	Patient KPIs, outcome distribution, readmission status and resource snapshot.
2	Patient Analytics	What patterns appear in patient records?	Patient metrics, outcome comparisons, length-of-stay distribution and cost by condition.
3	Resource Utilization	How are beds and staff allocated and used?	Bed occupancy, hospital bed comparisons and allocated versus on-duty staff.
4	Data Explorer	Can the figures be inspected and verified?	Dataset selection, previews, row and column counts, missing values and CSV export.

Shared Controls

The existing application provides sidebar navigation and patient filters for gender and condition, plus a reset button. These filters should affect patient-focused pages consistently. Resource records should not be filtered by patient attributes because the datasets do not have a verified common key.

Visual Design

Use the existing dark navy background, blue and teal chart accents, clear headings, KPI cards, readable labels and consistent spacing. Avoid unnecessary new pages or decorative charts.

2. Overview — Hospital Snapshot

Purpose: Give users a quick view of patient performance and resource status.

Component	Planned Visualization	Interpretation
Patient KPIs	Metric cards	Total patient records, average recorded length of stay, readmission rate and recovery rate.
Patient outcomes	Donut chart	Compare the recorded Recovered and Stable outcomes.
Readmission status	Bar chart	Compare records marked Yes and No.
Resource snapshot	Metric cards	Show allocated beds, occupied beds, aggregate occupancy and staff on-duty rate.

Design notes

Keep the highest-value metrics at the top and supporting charts below. Clearly distinguish patient-derived metrics from resource-derived metrics. The resource figures are sums across resource records and must not automatically be interpreted as unique hospital-wide totals.

3. Patient Analytics

Purpose: Explore differences in outcomes, recorded length of stay and patient costs.

Component	Visualization	Required Fields
Patient outcome comparison	Bar or donut chart	Outcome, patient record count
Length of stay	Histogram	Length_of_Stay
Average cost	KPI card	Cost
Average cost by condition	Horizontal bar chart	Condition, Cost
Patient filters	Sidebar multiselects	Gender, Condition

Interpretation cautions

The dataset contains 984 patient analysis records. The length-of-stay unit is not verified, so label it as recorded length of stay rather than assuming days. The dataset does not contain verified admission and discharge dates for a reliable time-series patient-flow analysis.

4. Resource Utilization

Purpose: Show how available bed and staffing allocations compare with usage.

Component	Visualization	Calculation / Field
Beds allocated	KPI card	Sum of beds_allocated
Beds occupied	KPI card	Sum of beds_occupied
Bed occupancy rate	KPI card	Total occupied beds / total allocated beds × 100
Hospital comparison	Grouped bar chart	beds_allocated and beds_occupied grouped by hospital_name
Staffing comparison	Bar chart	staff_allocated compared with staff_on_duty
Staff on-duty rate	KPI card	Total staff_on_duty / total staff_allocated × 100

Interpretation cautions

Some records have zero allocated beds or staff. Per-record rate calculations must handle division by zero. Aggregated rates should be described as rates across the available resource records until the dataset's record grain is confirmed.

5. Data Explorer and Interactions

Feature	Expected Behavior
Dataset selector	Switch between Patient Data, Resource Data and KPI Summary.
Dataset preview	Display a configurable sample of rows.
Quality indicators	Show row count, column count and missing-value count.
CSV export	Allow users to download the selected table.
View all records	Expand a full table when needed.
Sidebar navigation	Move between the four existing pages.
Patient filters	Filter patient analysis by gender and condition.
Reset filters	Restore the default filter selections.

Navigation Flow

Overview → Patient Analytics → Resource Utilization → Data Explorer. Users can also jump directly to any page through the sidebar.

6. KPI Reference and Data Limitations

KPI	Definition
Total Admissions	Count of patient analysis records; this is a record count, not verified unique admissions.
Average Length of Stay	Mean of Length_of_Stay; the measurement unit is unverified.
Readmission Rate	Records marked Yes / all eligible records × 100.
Average Patient Cost	Mean of Cost in the patient analysis dataset.
Recovery Rate	Records with Outcome = Recovered / all eligible records × 100.
Bed Occupancy Rate	Sum of beds_occupied / sum of beds_allocated × 100.
Staff On-Duty Rate	Sum of staff_on_duty / sum of staff_allocated × 100.

Known limitations

The current datasets do not provide a reliable shared key to link patient records to resource records. They do not support a verified department efficiency score, department-level comparisons, monthly admission trends, discharge tracking or patient movement analysis. These features require additional suitable data and must not be fabricated.

Quality and validation

Re-run the KPI generation and validation scripts after source data changes. Compare workbook totals with independently calculated values, check missing values and duplicates, and confirm that dashboard labels match KPI definitions.

Milestone 2 Deliverables

Deliverable	Status / Purpose
hospital_final_dataset.xlsx	Excel workbook containing prepared datasets and KPI sheets.
generate_hospital_kpis.py	Repeatable KPI calculation and workbook generation.
dashboard_storyboard.pdf	This dashboard planning and interaction document.

Deliverable	Status / Purpose
Streamlit app.py	Interactive dashboard prototype replacing the planned Tableau prototype.